

Response to Reviewer 1

General Comment. *This revised manuscript is much improved, however there are a number of issues still to be addressed prior to it being ready for publication.*

Response. We sincerely thank the comments and criticisms. We carefully considered each of the remarks and made substantial efforts to revise the manuscript accordingly.

Comment. *There are still typos such as 'lysine', 'contraction', 'insaturation'.*

Response: Thank you for pointing this out. Revised accordingly.

Page 8, line 169, 171: lysin=lysine

Page 14, line 282-284: contraction = compression

Page 27, line 520: insaturated = unsaturated

Comment. *The IR is an improvement, however I think there is more information to be gleaned from this part of the study and a few more details are required such as how long were b) and d) exposed to D₂O as this may affect the extent of H-D exchange? (For c the time is given as 10 min). This is important since as the authors state, there are varying extents of shift for the amide II. Can the authors fit the data to determine the proportion of residues exposed which may help understand the folded state of the protein?*

Response. Deuterium oxide was used to the only purpose to avoid the spectral absorption of H₂O, if any after drying, at 1650 cm⁻¹. The comment about the change in the amide II band with deuteration (page 7, line 145-147) was included just to inform that the hydrophobic peptides were actually in contact with the aqueous solvent. We consider that at this point of our study the kinetic aspects, that is, the time in contact with D₂O, are not important to describe the structure of the peptides, which is actually the reason for this experiment. Please take into account that spectral shape of amide I, which give us information about the secondary structure, does not suffer relevant changes upon deuteration.

We cannot evaluate the proportion of exposed residues from the present data. A careful measurement of deuterium exchange as a function of time under well controlled deuteration conditions could give us information about the conformational dynamics of the peptides. Most probably, for these relatively small peptides the amide proton exchange would be complete at long times precluding the distinction between exposed and occluded residues. We take the reviewers comment as a valuable advice to start a new project using H-D exchange to study the dynamics of the peptides inserted in a lipid bilayer environment.

The identification of the component bands by FSD and band fitting to estimate the proportion of secondary structure was included in the last version. The results are described in page 7 bottom, page 8 first paragraph. The experimental details and the table with bands and their proportions are in Supplementary Information.

Comment. *Figure 3 caption: Refer to legend for colors if not explained fully.*

Response. The figure caption was revised. The colour coding and values of hydrophobic moment were included in the helical wheels. Please see the new legend of Figure 3 in page 10.

Comment. *Consistency: hydrophobic moment and hydrophobic momentum are used interchangeably. I think it is the former.*

Response. Revised accordingly. Please see page 9, lines 180 and 183.

Comment. *Also, still the arrow size does not correspond to the values in the table. For example, β 3 IC has the largest arrow and the smallest (non-zero value). The values for the TM segments of the two peptides are virtually the same but the arrows differ in length.*

Response We have carefully checked the values in Table 1 and the arrows shown in Figure 3, and we do not find the mismatch between the arrow sizes and the corresponding values. The values of hydrophobic moments were now included in the wheel in Figure 3, please notice that they are consistent with the arrow sizes.

Comment. *was a bit annoyed, that despite pointing out the mismatch between the main text and the sequence I. The SI, the same problems persists and were not corrected. The text states that the theEC is residues 1-111 and the TM is 12-36 but the SI figure shows the brackets for the TM domain as 11-38. While not crucial to the discussion it shows a lack of attention to detail and to making corrections that have already been pointed out.*

Response: We are truly sorry for this mistake. We have revised **Table 1** in to make sure that the table is consistent with Supporting Information.

Comment. *I disagree that the peptides cause a complete loss of DPPC phases transition. It is still evident with 2% and 3.5% in the isotherms and as a small minimum in the compressibility plots.*

Response: We agree with the reviewer observation. We changed the sentence as follows: “A major effect of both peptides was a marked reduction in the DPPC phase transition, observable even at low peptide proportions.” Please see page 14, line 278.

Comment. *I would adjust the co toast on the BAM images to see if there is any evidence of the transition I. The background that is obscured by the brightness of the peptide domains,*

Similarly, I think, if you look carefully and adjust the contrast, you will see that $\beta 3$ also shows the round patches with lower contrast. I think May also be present at 15 mNm⁻¹ for αIIb .

Response: We appreciate the advice. The contrast was increased in the panels of Figure 10. In spite of obtaining a better figure, we find essentially the same information as in the previous version.

Comment. *At the start of the BAM section, please give the rationale for using 0.005%peptide which matches none of the other experiments about the extrapolation from 2 points to determine a thickness for the peptides.*

Response. The explanation was included in page 22 lines 433-435. Please take into account that the estimation of the thickness in the BAM experiments is independent of the other, classic monolayer experiments.

Comment. *Additionally, how does the $\beta 3$ extrapolate to 110 at 30 mNm⁻¹ when the plot x-axis goes to 35 mNm⁻¹ and the value is less than 100?*

Response. The right number is 90 mN.m⁻¹. It was corrected in the manuscript. Please see page 24 line 468.

Response to Reviewer 3

Comment. *The difficulties to infer relevant features regarding lipid-protein interactions of transmembrane protein regions from monolayer experiments have been now properly discussed. This enhances the validity of the study and extends its potential utility as an example of an approach applicable in other systems. From this point of view, the study has gained relevance and deserves to be seen by scientists in the field.*

Response. We greatly appreciate this encouraging feedback. We are glad to hear that the revised discussion helped clarify the challenges and relevance of using monolayer experiments to study transmembrane protein–lipid interactions. We appreciate the recognition of the broader applicability of our approach.

Comment. *The structure of the paper has improved in my opinion by separating Results and Discussion. Still, a “Results and Discussion” section appears in the text, I guess as a mistake.*

Response: We appreciate the advice. The mistake in the section title was corrected.

Comment. *The discussion regarding the compressibility of the peptide networks beyond the compressibility of the individual molecules is very pertinent and relevant. When the authors refer to “connectivity is required to transmit the compressibility of the phase to the whole system”, they are actually referring to the **percolation threshold**, a well-established concept describing the dependence of the general properties of a film/membrane on the distribution of (dominant) phases. Mentioning this could enhance the impact of the discussion.*

Response. We thank the valuable suggestion. We agree that the connectivity concept we refer to aligns with the notion of a percolation threshold. We have revised the paragraph to include this term explicitly. Please new paragraph and citation in top of page 28.